

Fall 2025 (subject to change)

Instructor: Hamed Mamani	
Office: PCAR 547	Office Hours: By appointment
Phone: 206-543-0787	Class Schedule: M 6-9:15pm
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Course Description and Objectives

Data analytics is a set of techniques that enterprises use to gain insight from their data and make better decisions. Many firms in a variety of industries use these techniques: Google, Amazon, Target, Coca-Cola, WalMart, Capital One. These techniques are also applicable to the many functional areas of business, such as operations, marketing, accounting, finance, etc. Furthermore, the modern abundance of data, so-called “Big Data,” underscores the value that analytics can provide a firm, be it non-profit, for-profit, or government.

This course introduces data analytic techniques via quantitative tools and sophisticated software (R, Radiant and Tableau). These techniques are drawn from machine learning, data mining, and optimization. Note that this is not a technical or theoretical course. This course does not aim to produce experts in statistical analysis; rather, the aim is to provide students competency to interact with and manage a team of analytics professionals. Furthermore, this is not a technical or theoretical course; we will instead focus on the application of analytics techniques to real business situations, with the aim of creating insight and value.

The course is divided into the following six main modules under three topic areas of Descriptive Analytics, Predictive Analytics, and Prescriptive Analytics:

1. Supervised Learning: Prediction (Predictive Analytics)
2. Supervised Learning: Classification (Predictive Analytics)
3. Visualization (Descriptive Analytics)
4. Unsupervised Learning: Clustering (Descriptive and Predictive Analytics)
5. Data Preprocessing

6. Team Projects

Our course goals are the following:

1. Students should be able to think critically about data analysis, which includes selecting the right type of analysis for a given task.
2. Students should be able to identify opportunities of applying data analytics, in real business settings.
3. Students should be well equipped to become data-savvy managers.

To achieve the above goals, lectures will cover the major concepts and analytical tools. Cases and practice problems will allow you to analyze different industry settings, analyze different company strategic problems, and identify key issues related to data and modeling.

Recommended Textbook (not required)

The Analytics Edge, Bertsimas, O'Hair, Pulleyblank. Dynamic Ideas. 1st Edition; 2016.

Computer Software

Students are expected to be familiar with using Microsoft Excel software (e.g., modules 1-3 on the Excel for Business online course, available on canvas.uw.edu. In addition, we will be using Tableau, R, and Radiant in this class. These software packages are both available in the computer labs in Paccar 190. Student licenses to Tableau will be provided; R and Radiant are free.

- Radiant installation instructions can be found here: <https://radiant-rstats.github.io/docs/install.html> Links to an external site.
- Tableau can be downloaded at <http://www.tableau.com/products/trial> Links to an external site.

Assignments

Most weeks a case will be assigned. Students are expected to work either individually or in groups on the cases. However, each student must submit his/her own solution. These assignments are due at the beginning of class on their posted due date and are electronically submitted through the class website on canvas.uw.edu.

Group Project

The final assignment of this course will consist of a group project where teams (of 4-5 students) will apply the techniques of the course to real data. There are four sets of deliverables:

1. Project proposal, due November 3.
2. A 15-20 minute presentation to the class on December 1.
3. A 5-10 page writeup, due December 7.
4. Peer evaluations of your group members.

Further details will be discussed in class.

Grading

Student grades will be calculated applying the following allocation in a simple weighted average:

- | | |
|-------------------|-----|
| 1. Assignments | 50% |
| 2. Group Project. | 40% |
| 3. Quizzes | 10% |

The (required) target median GPA in this class is 3.4 – 3.6.

Course Website

All course materials will be distributed electronically through the website canvas.uw.edu. Assignments and Team Projects are submitted electronically through this website (by their due dates!).

CLASS SCHEDULE AND TOPICS

(subject to revision)

Session 1:

Introduction to Data Analytics and R/Radiant

Linear Regression Model Overview

Session 2:

Predictive Analytics:

Prediction: Linear Regression (Supervised Learning)

- Predicting Wine Quality

Session 3: No Class

Session 4:

Predictive Analytics:

Classification: Logistic Regression (Supervised Learning)

- Predicting Quality in Healthcare

Linear Regression Assignment Due

Session 5:

Descriptive Analytics:

Clustering (Unsupervised Learning)

- Netflix and Recommendation Systems
- Wine Clustering

Logistic Regression Assignment Due

Session 6:

Descriptive Analytics:

Visualization and Story Telling with Data I

Clustering Assignment Due

Session 7:

Descriptive Analytics:

Visualization and Story Telling with Data II

Visualization I Assignment Due

Team Project Proposals

Session 8:

Data Pre-Processing:

Data Cleaning, Manipulation and Preprocessing

Clustering using Tableau

Visualization II Assignment Due

Session 9:

Qualitative Prediction Methods

Future Trends in Data Analytics

Session 10:

Team Project Presentations